# A Theorem Which Is Not Common Knowledge in an Untyped Theory of Knowledge

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**Abstract**

Gorbow's theory KT combines an arithmetized knowledge predicate, compositional truth, veracity and theorem-only necessitation. Common knowledge is represented by a self-referential fixed point. Gorbow asks whether the common-belief necessitation theorem for the subsystem DCB extends to KT. We give a negative answer for the published theory with base PA in the expanded language. There is a truth-free sentence $\theta$ such that KT proves $\theta$ and proves the negation of its common-knowledge assertion, although every fixed finite iteration of everyone-knows applied to $\theta$ is a KT theorem. More generally, for each permitted selector, a local-consistency fixed point is equivalent in DCB to everyone selected knowing a contradiction. In KT it is equivalent to the selected group being empty. We prove this classification and supply a direct one-agent revision argument for consistency of KT and KT with the untyped Barcan principle. All necessitation rules are used only on theorems.

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## 1 The common-knowledge necessitation question

Ordinary necessitation says that once a sentence is a theorem, one may infer that an agent knows it. Common-knowledge necessitation would allow a further inference: every selected agent knows the sentence, knows that every selected agent knows it, and so on in the sense formalized by the common-knowledge predicate.

In a typed modal system these principles often fit together naturally. Gorbow's setting [1] is deliberately untyped. Knowledge and truth apply to codes of sentences in the full language, including sentences which themselves discuss knowledge and truth. Common knowledge is expressed by a diagonal fixed point inside arithmetic. These choices permit useful self-reference but make the passage from individual to common necessitation a substantive theorem, rather than a matter of notation.

The common-belief subsystem DCB has a theorem-to-common-belief rule. In his concluding discussion, item 5(d), Gorbow asks how its results extend to the larger knowledge theory KT, including the corresponding common-knowledge rule. We show that the unrestricted extension fails. The counterexample is particularly sharp: the proposed common knowledge is not merely unproved; its negation is a theorem.

The result concerns the published full-language rule. An arithmetic-only rule, a different fixed-point constructor, or an additional restriction on selectors would require a separate analysis.

## 2 The source theories and coding conventions

Work in Gorbow's expanded first-order language $L$. It contains the language of arithmetic, an agent predicate $\mathsf{Ag}$, a binary knowledge predicate $\mathsf{K}(a, u)$, a unary knowledge predicate for all agents, a truth predicate $\mathsf{T}(u)$, and standard definitional syntax coding. The sublanguage $L_-$ consists of arithmetic, the source's universal coding relation and function $U, u$, and $\mathsf{Ag}$; it excludes both knowledge and truth. Its interpretations will remain fixed in the consistency construction. The base in our consistency theorem is exactly $\mathsf{PA}(L)$ with the source's definitional syntax apparatus. In particular induction is available for formulas in $L$. No arbitrary extra base theory is assumed.

For a literal sentence $\sigma$, $\ulcorner\sigma\urcorner$ abbreviates its numeral code. Codes, substitution functions and the numeral denoting a code must be distinguished: $\mathsf{CK}_A(\ulcorner\sigma\urcorner)$ is a sentence and has its own code $\ulcorner\mathsf{CK}_A(\ulcorner\sigma\urcorner)\urcorner$.

Put $D = \mathsf{DCB}$. Its relevant axioms, with the source sentencehood guards understood, are

$$\exists a\, \mathsf{Ag}(a), \tag{2.1}$$

$$\mathsf{K}(a, \ulcorner\sigma \to \tau\urcorner) \wedge \mathsf{K}(a, \ulcorner\sigma\urcorner) \;\longrightarrow\; \mathsf{K}(a, \ulcorner\tau\urcorner) \quad (\mathsf{Ag}(a)), \tag{2.2}$$

$$\mathsf{Pr}_D(\ulcorner\sigma\urcorner) \;\longrightarrow\; \mathsf{K}(a, \ulcorner\sigma\urcorner) \quad (\mathsf{Ag}(a)). \tag{2.3}$$

These are instances of universally coded axioms, not restrictions of their domains to an externally chosen finite collection. The unary knowledge predicate is linked by the source bridge to $\forall a(\mathsf{Ag}(a) \to \mathsf{K}(a, u))$.

The index of $\mathsf{Pr}_D$ in (2.3) is $D$ itself. It is not the provability predicate for KT. Gorbow constructs its recursive presentation by a syntactic fixed point. We use that presentation and the ordinary derivability conditions. In particular Löb's theorem applies to $D$: if

$$D \vdash \mathsf{Pr}_D(\ulcorner\delta\urcorner) \to \delta, \quad \text{then} \quad D \vdash \delta.$$

This is the standard metatheorem for provability, not an extra knowledge axiom. Its application does not require a consistency assumption.

The theory $\mathsf{FS}$ adds uniform compositional truth axioms for the fixed $L_-$ atoms, Boolean connectives and quantifiers. The atomic clauses apply to closed arithmetic term instances; the

quantified clauses use closed $L$-term substitution. It also has the two theorem rules

$$\frac{\vdash \sigma}{\vdash \mathsf{T}(\ulcorner\sigma\urcorner)}, \qquad \frac{\vdash \mathsf{T}(\ulcorner\sigma\urcorner)}{\vdash \sigma}.$$

For example the Boolean clauses include $\mathsf{T}(\ulcorner\neg\sigma\urcorner) \leftrightarrow \neg\mathsf{T}(\ulcorner\sigma\urcorner)$ and $\mathsf{T}(\ulcorner\sigma\wedge\tau\urcorner) \leftrightarrow (\mathsf{T}(\ulcorner\sigma\urcorner)\wedge\mathsf{T}(\ulcorner\tau\urcorner))$. Their full coded versions, not just these literal instances, are part of $\mathsf{FS}$.

Finally KT extends $D$ and $\mathsf{FS}$ by veracity and theorem-only knowledge necessitation:

$$\mathsf{Ag}(a) \to (\mathsf{K}(a,\ulcorner\sigma\urcorner) \to \mathsf{T}(\ulcorner\sigma\urcorner)), \qquad \frac{\vdash \sigma}{\vdash \forall a(\mathsf{Ag}(a) \to \mathsf{K}(a,\ulcorner\sigma\urcorner))}. \tag{2.4}$$

No rule is licensed inside a derivation under temporary assumptions. The source's published figures give the full syntactic schemata; the displayed formulas isolate exactly the instances used below.

### 2.1 Selected agents and common knowledge

Let $A(x)$ be any permitted $L$-formula with $x$ its only free variable. For a literal sentence $\sigma$ define

$$E_A(\sigma) = \forall x\big(\mathsf{Ag}(x) \to (A(x) \to \mathsf{K}(x,\ulcorner\sigma\urcorner))\big), \qquad N_A = \exists x(\mathsf{Ag}(x) \wedge A(x)).$$

The formula $A$ may contain knowledge or truth; no external evaluation of it is assumed.

We use Gorbow's chosen fixed-point formula $\mathsf{CK}_A$. Its defining equivalence, after numeral substitution, gives for each literal $\sigma$ the $D$ theorem

$$\mathsf{CK}_A(\ulcorner\sigma\urcorner) \;\leftrightarrow\; E_A(\sigma) \wedge E_A\big(\mathsf{CK}_A(\ulcorner\sigma\urcorner)\big). \tag{2.5}$$

The second conjunct therefore involves the code of the actual common-knowledge sentence. We do not replace it by an informal infinite conjunction or by a different predicate satisfying an external equation.

**Lemma 2.1.** *For fixed literal sentences $\sigma, \tau$:*

1. *if $D \vdash \sigma$, then $D \vdash E_A(\sigma)$;*
2. *$D \vdash E_A(\sigma \to \tau) \to (E_A(\sigma) \to E_A(\tau))$;*
3. *$D \vdash \mathsf{Pr}_D(\ulcorner\sigma\urcorner) \to E_A(\sigma)$.*

*Proof.* For the first assertion, a fixed actual proof of $\sigma$ has its provability certified by the arithmetic base. Apply (2.3) and restrict to selected agents. The second assertion is pointwise (2.2) followed by first-order logic. The third is (2.3) with the same restriction. No distribution through a quantified knowledge claim is used. □

## 3 The local-consistency fixed point collapses

Let $b$ be the arithmetic contradiction $0 = S0$, and set

$$\theta_A = (E_A(b) \to b), \qquad c_A = \mathsf{CK}_A(\ulcorner\theta_A\urcorner).$$

The sentence $\theta_A$ says that if all selected agents know the contradiction, the contradiction holds. It has the form of a local consistency assertion for the selected knowledge predicates.

**Theorem 3.1.** *For every allowed selector $A$,*

$$D \vdash c_A \leftrightarrow E_A(b).$$

*Proof.* Write $\delta = (c_A \to E_A(b))$. Reason in ordinary first-order deduction over $D$ under temporary assumptions $\mathsf{Pr}_D(\ulcorner \delta \urcorner)$ and $c_A$. Lemma 2.1(3) yields $E_A(\delta)$, while (2.5) gives $E_A(c_A)$ and $E_A(\theta_A)$. Two applications of Lemma 2.1(2) give, successively,

$$E_A(E_A(b)), \qquad E_A(b).$$

The first uses $\delta = (c_A \to E_A(b))$; the second uses $\theta_A = (E_A(b) \to b)$. Discharging the assumptions proves $D \vdash \mathsf{Pr}_D(\ulcorner \delta \urcorner) \to \delta$. Löb's theorem now gives $D \vdash \delta$, proving the forward implication. No necessitation rule has been applied to an assumption.

For the converse, $D$ proves the two tautological implications $b \to \theta_A$ and $b \to c_A$. Lemma 2.1(1),(2) gives

$$E_A(b) \to E_A(\theta_A), \qquad E_A(b) \to E_A(c_A).$$

Their conjunction and (2.5) imply $E_A(b) \to c_A$. □

**Corollary 3.2.** *The same fixed point is $D$-equivalent to common belief of the contradiction: $D \vdash c_A \leftrightarrow \mathsf{CK}_A(\ulcorner b \urcorner)$.*

*Proof.* The defining equation for $\mathsf{CK}_A(\ulcorner b \urcorner)$ gives its implication to $E_A(b)$. Conversely the $D$ theorem $b \to \mathsf{CK}_A(\ulcorner b \urcorner)$, placed under $E_A$, gives $E_A(b) \to E_A(\mathsf{CK}_A(\ulcorner b \urcorner))$. Use the defining equation again, then Theorem 3.1. □

## 4 Veracity gives the exact selector classification

**Theorem 4.1.** *In the original KT theory,*

$$\mathsf{KT} \vdash \theta_A \leftrightarrow N_A, \qquad \mathsf{KT} \vdash c_A \leftrightarrow \neg N_A.$$

*Proof.* The atomic truth clause for the closed equality $b$ gives $\mathsf{T}(\ulcorner b \urcorner) \leftrightarrow b$. Since PA proves $\neg b$, veracity proves $\forall x(\mathsf{Ag}(x) \to \neg \mathsf{K}(x, \ulcorner b \urcorner))$. First-order logic therefore gives $E_A(b) \leftrightarrow \neg N_A$. This is valid even when $A$ contains knowledge or truth: no assumption about its extension is needed. With $\neg b$, the implication $E_A(b) \to b$ is equivalent to $\neg E_A(b)$, hence to $N_A$. The second equivalence follows from Theorem 3.1. □

Take $A(x)$ to be $x = x$, denoted $\top$. Nonemptiness of agents gives the explicit truth-free counterexample

$$\theta = \big(\forall x(\mathsf{Ag}(x) \to \mathsf{K}(x, \ulcorner b \urcorner)) \to b\big), \qquad \mathsf{KT} \vdash \theta, \quad \mathsf{KT} \vdash \neg \mathsf{CK}_\top(\ulcorner \theta \urcorner). \tag{4.1}$$

Thus adding unrestricted common-knowledge necessitation to KT makes it inconsistent. Once consistency of KT is established, the displayed negation also yields nonprovability of the common-knowledge assertion in KT itself.

**Corollary 4.2.** *For every fixed external $r \in \mathbb{N}$, $\mathsf{KT} \vdash E_\top^r(\theta)$, where $E_\top^0(\theta) = \theta$ and $E_\top^{r+1}(\theta) = E_\top(E_\top^r(\theta))$.*

*Proof.* Starting from (4.1), apply theorem-only knowledge necessitation $r$ times. Each premise is an already established theorem. □

This is a separate theorem for every standard finite $r$. It does not assert a formula internally quantifying over iteration lengths. In particular it supplies no inference from all finite iterations to the chosen self-referential common-knowledge sentence. That missing inference is exactly what the example exposes.

The empty-selector case is a useful check. If KT proves $\neg N_A$, it proves $c_A$ and $\neg\theta_A$. If the nonemptiness of the selected group is undecided, the two equivalences of Theorem 4.1, rather than an externally chosen case, are the result.

## 5 A direct consistency argument

Gorbow proves consistency by generalized revision semantics [1, Theorem 15 and Corollary 16]. We include a direct one-agent specialization, both to make the nonprovability conclusion available here and to explain why the preceding results coexist. The construction uses standard external satisfaction. It does not provide one standard structure satisfying every KT theorem.

**Theorem 5.1.** *The source theories KT and* $\mathsf{KT} + \mathsf{UBF}$ *are consistent when their base is* $\mathsf{PA}(L)$ *with its standard definitional syntax apparatus. Here* $\mathsf{UBF}$ *is the source untyped Barcan principle for universally known closed-arithmetic-term instances.*

*Proof.* Fix an $L_-$ expansion $M$ of the standard natural numbers satisfying the defining arithmetic syntax axioms. Interpret $\mathsf{Ag}$ by $\{0\}$, and keep all functions, syntax devices and $L_-$ relations fixed. Every instance of full $L$-induction holds in every expansion of the standard natural numbers, whatever interpretations are later given to knowledge and truth.

Let $P_0$ be the set of codes of literal $D$-theorems. This set exists without an initial consistency assumption. Define $M_n$ to have truth extension $P_n$, binary knowledge extension $\{(0,u) : u \in P_n\}$, and unary all-agents knowledge extension $P_n$. Recursively set

$$P_{n+1} = \{\ulcorner\sigma\urcorner : M_n \models \sigma\},$$

where $\sigma$ ranges over actual $L$-sentences. At each standard finite stage,

$$M_{n+1} \models \mathsf{T}(\ulcorner\sigma\urcorner) \quad\Longleftrightarrow\quad M_n \models \sigma. \tag{5.1}$$

The same equivalence holds for $\mathsf{K}(0,\ulcorner\sigma\urcorner)$ and for unary knowledge. Satisfaction on the right is external classical satisfaction.

*DCB holds at every stage.* At stage zero the theorem set is closed under modus ponens, so knowledge distribution holds. The provability predicate has its standard interpretation, so (2.3) holds. Nonemptiness and the unary/binary bridge hold by construction; the arithmetic base holds as just explained. Thus $M_0 \models D$. If $M_n \models D$, ordinary first-order soundness says every $D$-theorem is true in $M_n$, hence its code is in $P_{n+1}$. This gives (2.3) at the next stage. Classical truth in $M_n$ is closed under modus ponens, giving distribution there too. Induction proves $M_n \models D$ for all $n$. The soundness step concerns the ordinary first-order axiom theory $D$; it does not insert KT's additional theorem rules into $D$.

*Truth composition holds at positive stages.* For $n \geq 1$, membership in $P_n$ means truth in $M_{n-1}$. Classical satisfaction immediately gives all Boolean truth clauses in $M_n$, including for sentences

containing knowledge or truth. The $L_-$ atomic clauses hold because their relations and closed arithmetic term values remain fixed. For quantifiers, the substitution lemma applies to the fixed function interpretations. Closed $L$-terms denote natural numbers, and every natural number has a numeral. Thus truth of all closed-term instances in $M_{n-1}$ is exactly universal satisfaction there; the existential clause is similar. This verifies the full coded quantified clauses, since the ambient arithmetic universe is standard. Veracity holds even at stage zero, because the single agent's knowledge extension equals the truth extension. It does not assert that $P_n$ is the set of sentences true in $M_n$.

Call a sentence *eventually valid* when it is true in all $M_n$ beyond some standard finite stage depending on the sentence. Every $D$ axiom has threshold zero; every FS axiom has threshold at most one; veracity has threshold zero. Logical consequences of finitely many eventually valid sentences are eventually valid, by ordinary soundness in each sufficiently late structure.

If $\sigma$ is eventually valid, (5.1) makes $\mathsf{T}(\ulcorner\sigma\urcorner)$ eventually valid, and also gives eventual validity of $\forall a(\mathsf{Ag}(a) \to \mathsf{K}(a, \ulcorner\sigma\urcorner))$. Conversely eventual validity of $\mathsf{T}(\ulcorner\sigma\urcorner)$ implies eventual validity of $\sigma$ by shifting the stage back one. All three theorem rules preserve eventual validity. Induction over an actual finite KT proof therefore makes its conclusion eventually valid. The arithmetic falsehood $b$ is false at every stage, so it cannot be a theorem.

For the additional Barcan axiom, take $n \geq 1$. If the sole agent knows every closed arithmetic term instance of $\varphi(v)$ at stage $n$, then $M_{n-1}$ satisfies all these instances. Numerals cover the standard universe, so $M_{n-1} \models \forall v\varphi(v)$. Equation (5.1) says the agent knows that universal sentence at stage $n$. Thus the actual full-language UBF schema is eventually valid. The same proof induction establishes consistency of the extension separately. □

The proof uses no internally defined truth iteration, no common knowledge converse principle, and no assertion that finite revision stages stabilize globally. Its conclusion is external consistency, not an internal proof of $\mathrm{Con}(\mathsf{KT})$.

## 6 What the negative answer establishes

Combining Theorem 5.1 with (4.1) gives

$$\mathsf{KT} \nvdash \mathsf{CK}_\top(\ulcorner\theta\urcorner), \qquad \mathsf{KT} + \mathsf{UBF} \nvdash \mathsf{CK}_\top(\ulcorner\theta\urcorner).$$

Both theories nevertheless prove $\theta$ and the negation of that common-knowledge sentence. This answers the unrestricted common-knowledge necessitation question in the specified source case.

The selector theorem is slightly more informative than one counterexample: it identifies the exact role of nonemptiness. The DCB collapse also explains the obstruction structurally. A fixed point claiming common belief of local consistency allows provability in DCB to trigger Löb's theorem; it collapses to common belief of the contradiction. Adding veracity then turns that condition into emptiness of the selected group.

Strengthening the base by arbitrary axioms preserves the displayed syntactic derivations, but the consistency proof does not automatically extend. Likewise, alternative restricted necessitation rules are not classified by this counterexample. These boundaries keep the result aligned with the full-language question actually answered.

**Preparation.** This AI-assisted hand-proof manuscript includes the local fixed-point, selector and consistency arguments. The source theories and Löb's metatheorem are credited mathematical inputs. No formal certification or established priority is claimed.

## References


[1] P. Gorbow, *A genuinely untyped explanation of common belief and knowledge*, The Review of Symbolic Logic 18 (2025), no. 1, 160–185. doi:10.1017/S175502032400025X. We use the published DCB/FS/KT axioms, common-belief fixed point, Theorem 2, Theorem 15, Corollary 16 and concluding discussion 5(d).

[2] M. H. Löb, *Solution of a problem of Leon Henkin*, Journal of Symbolic Logic 20 (1955), no. 2, 115–118.